DELTA-SIGMA ADC ASIC DEVELOPMENT

Team 76

System Specifications

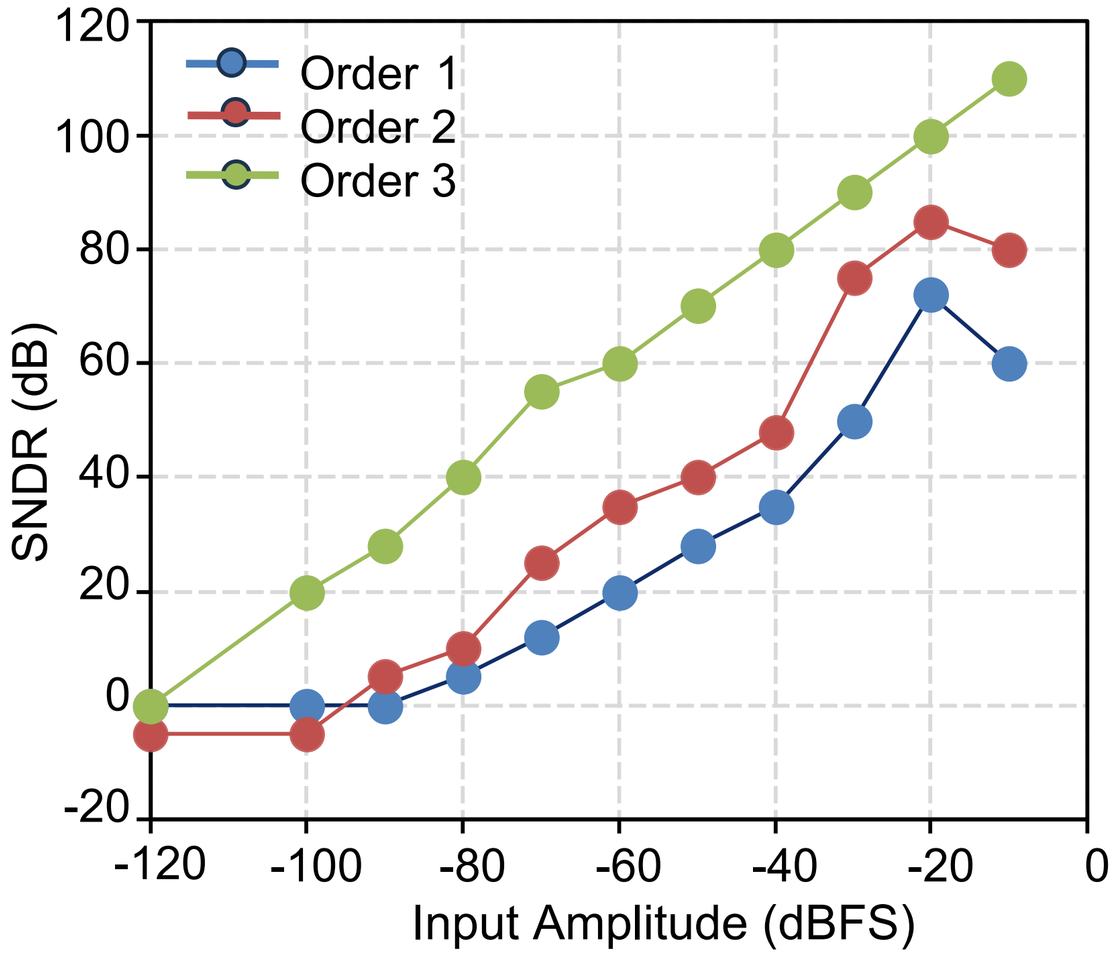
Parameter	Value
Over-Sampling Frequency (fs)	**4096 kHz**
Nyquist Sampling frequencies	**2 Modes - 1 kHz / 2kHz**
ADC Resolution	**20 bits**
ENOB Achieved	**18.5**
Technology used	**SCL 180 nm**
DSM Type	**3rd order CT-CT-DT**

Modulator NTF

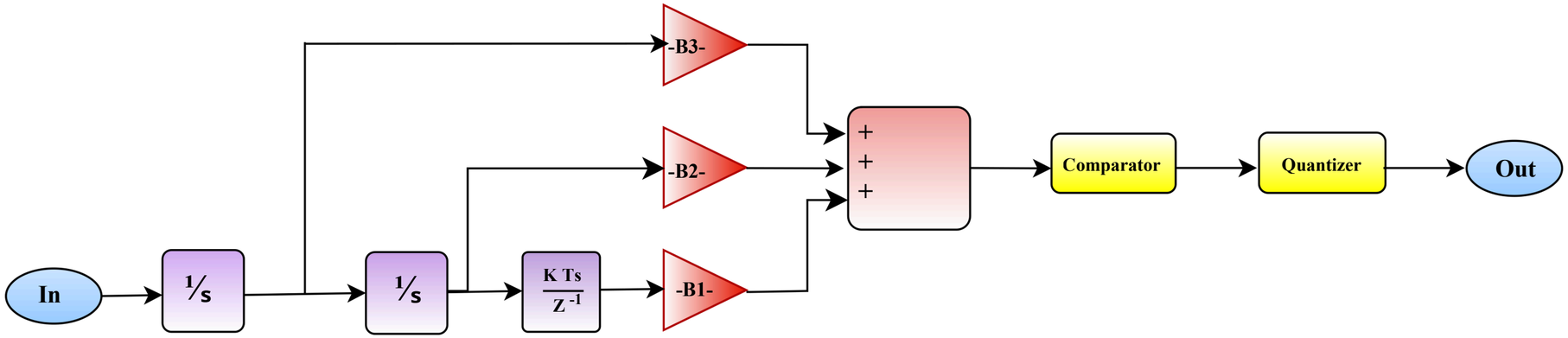
SNR simulation results of synthesised NTFs for various orders of modulators

Synthesised NTF of 3rd order modulator with OBG of 1.5

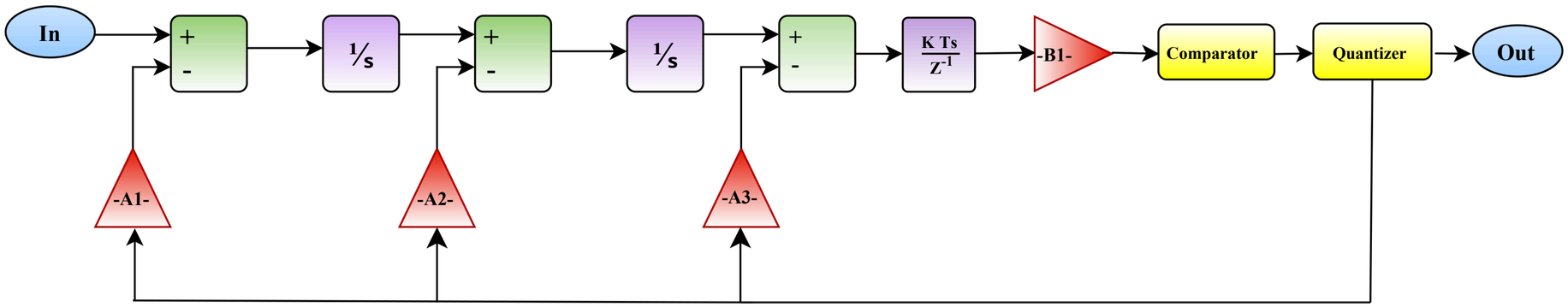
$$\mathrm{NTF}(z) = \frac{(z-1)^3}{(z-0.6694)\left(z^2-1.531z+0.6639\right)}$$



CIFB vs CIFF Modulators Structure

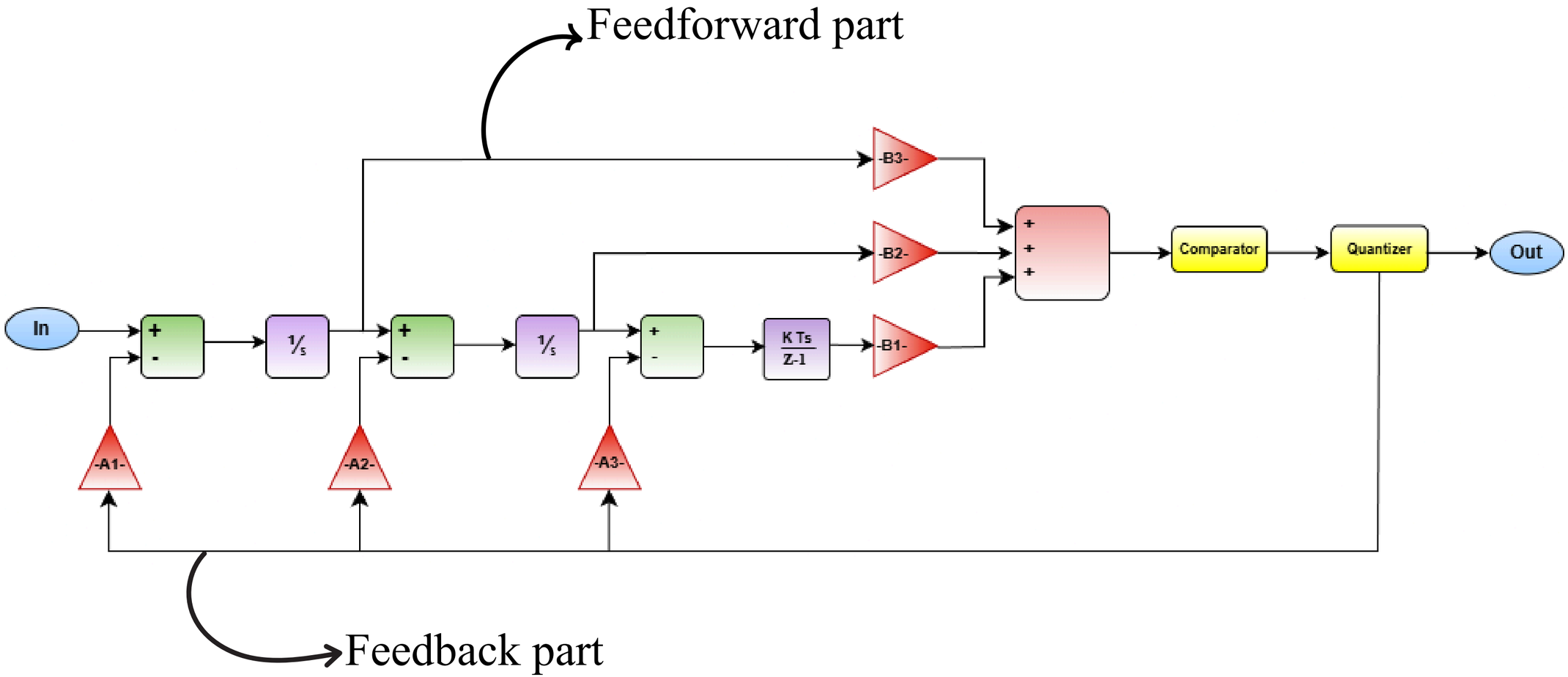


CIFF (Cascaded Integrator Feedforward Architecture)



CIFB (Cascaded Integrator Feedback Architecture)

Final modulator design: CIFB + CIFF



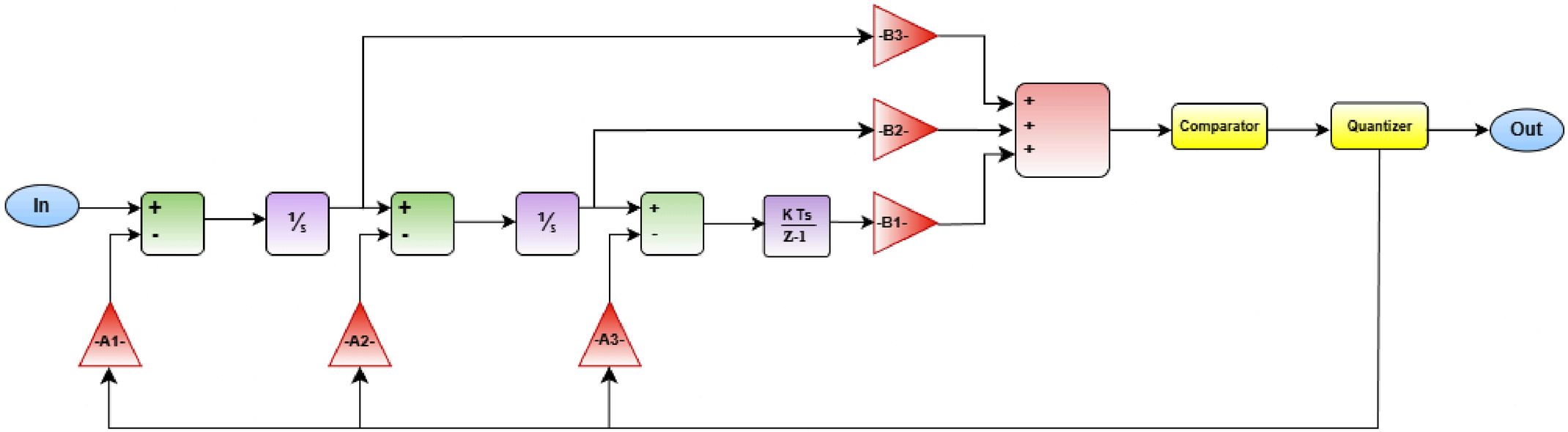
NTF of Final modulator design

$$\mathrm{NTF}(z) = \frac{(z-1)^3}{D(z)}$$

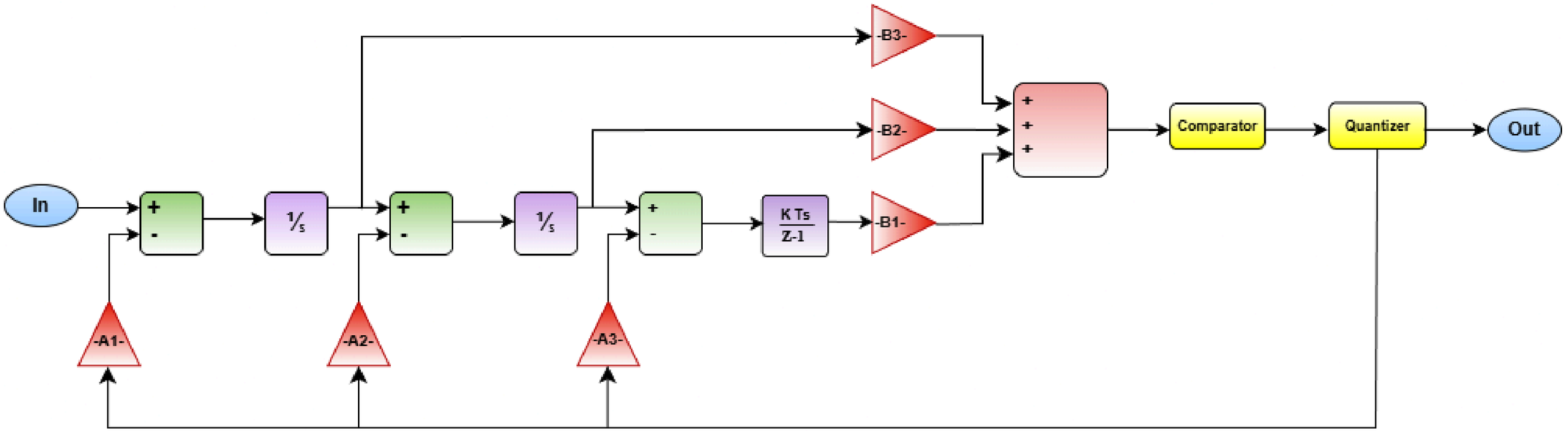
where,

$$D(z) = z^3 - \left(3 + A_1B_3 - A_2B_2 - A_3B_1\right)z^2 + \left(3 - A_1B_2 + A_2B_1 + 2A_1B_3 - 2A_2B_2 - 2A_3B_1\right)z + \left(1 + A_1B_2 - A_2B_1 - A_1B_1 - A_1B_3 + A_2B_2 + A_3B_1\right)$$

NTF of Third-Order Delta-Sigma Modulator



Why CT-CT-DT?



Continuous-time: Loop filter acts as an AAF, suppressing blockers before sampling

Discrete-time: High accuracy, no RC tuning issues

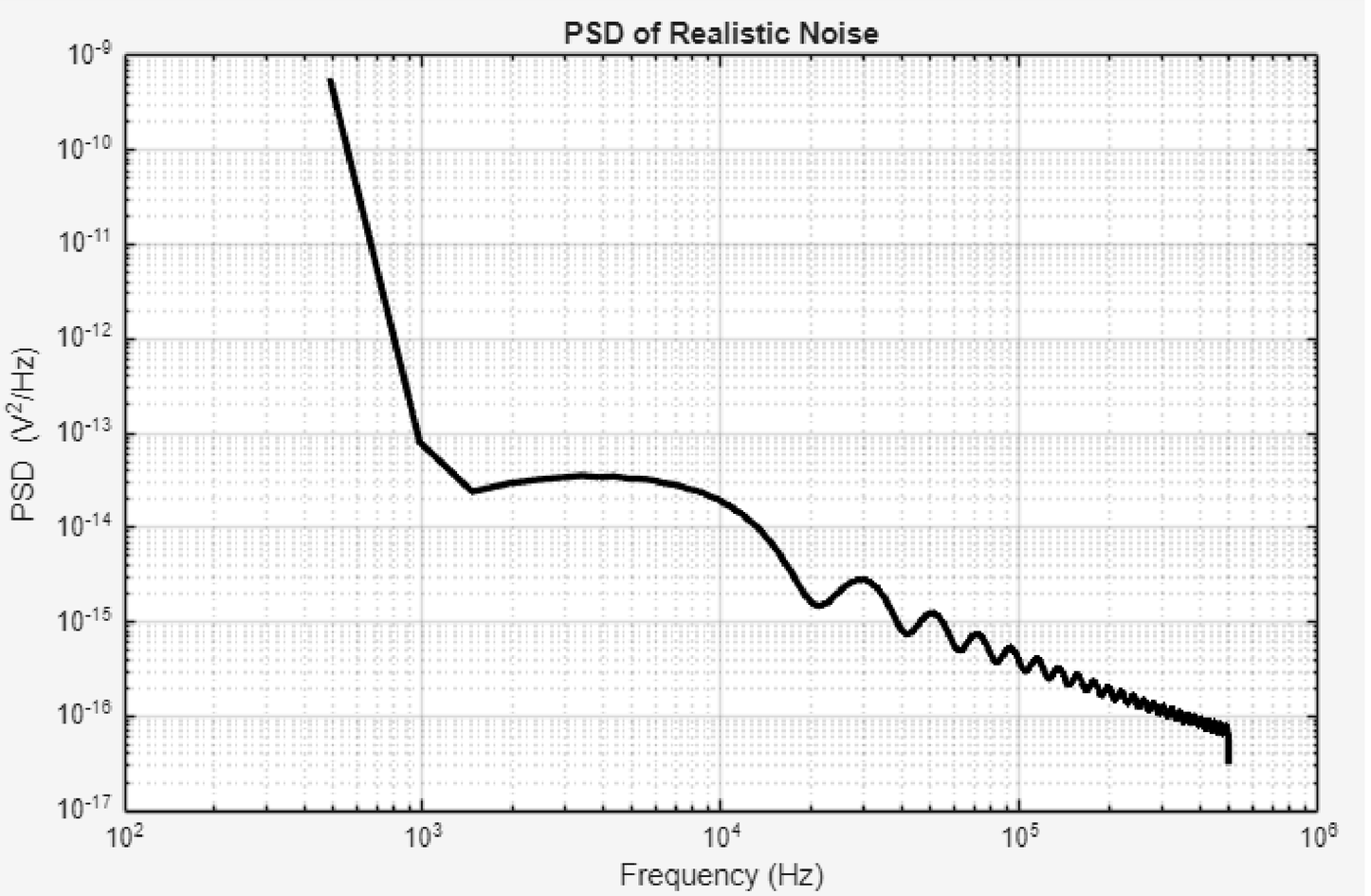
Noise PSD

$$S_{v_n}(f) = \frac{4kT\gamma}{g_m} + \frac{K_f}{C_{ox}^2 W L f^{\alpha}}$$

PSD $\propto$ 1/f $\rightarrow$ dominant at the desired bandwidth(0.25-1kHz)

Limits ENOB at high precision ADCs

Effective mitigation technique is required

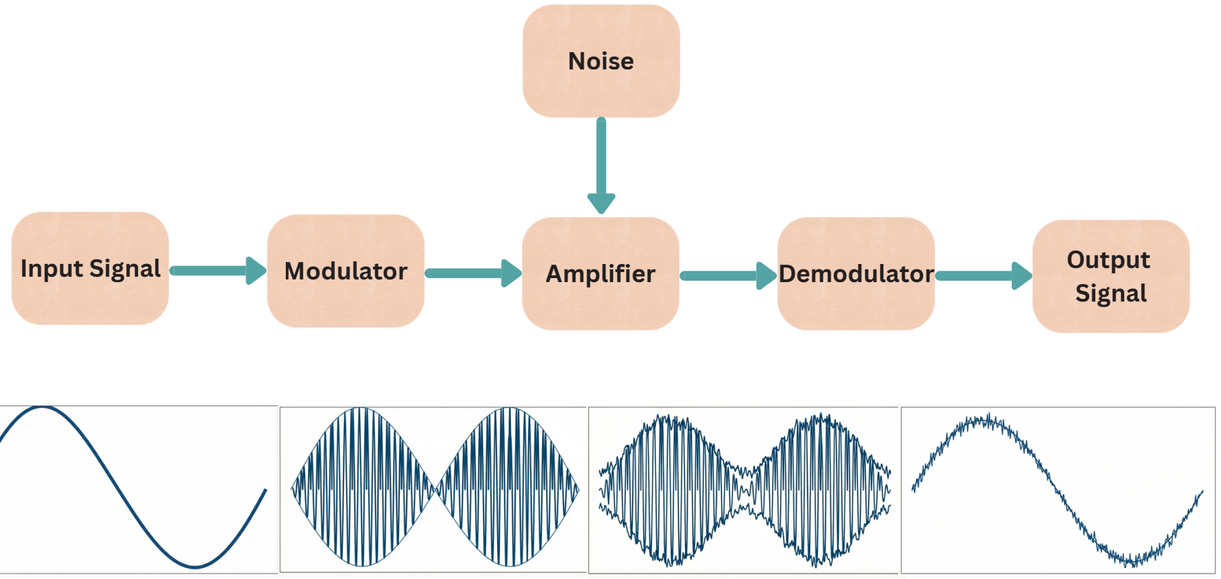


Modelled Flicker Noise

Noise Mitigation Techniques

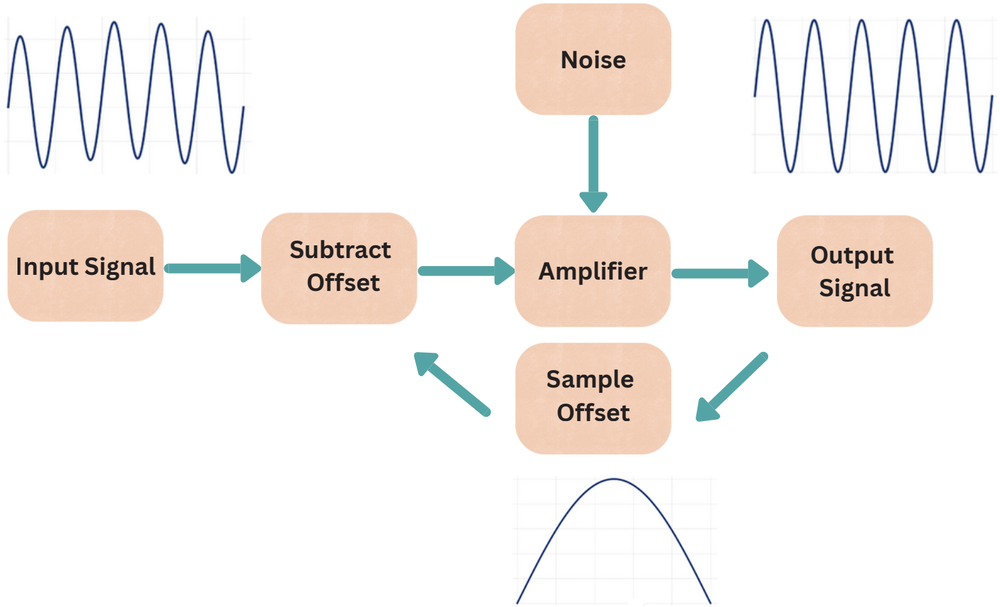
Chopping Circuit

- Modulates out the 1/f noise and demodulates to restore a cleaner signal.
- Creates ripple artifacts that require additional filtering.



Autozeroing

- Cancels offset and 1/f noise through sample and subtract operation.
- Introduces aliasing of broadband noise into the signal band.

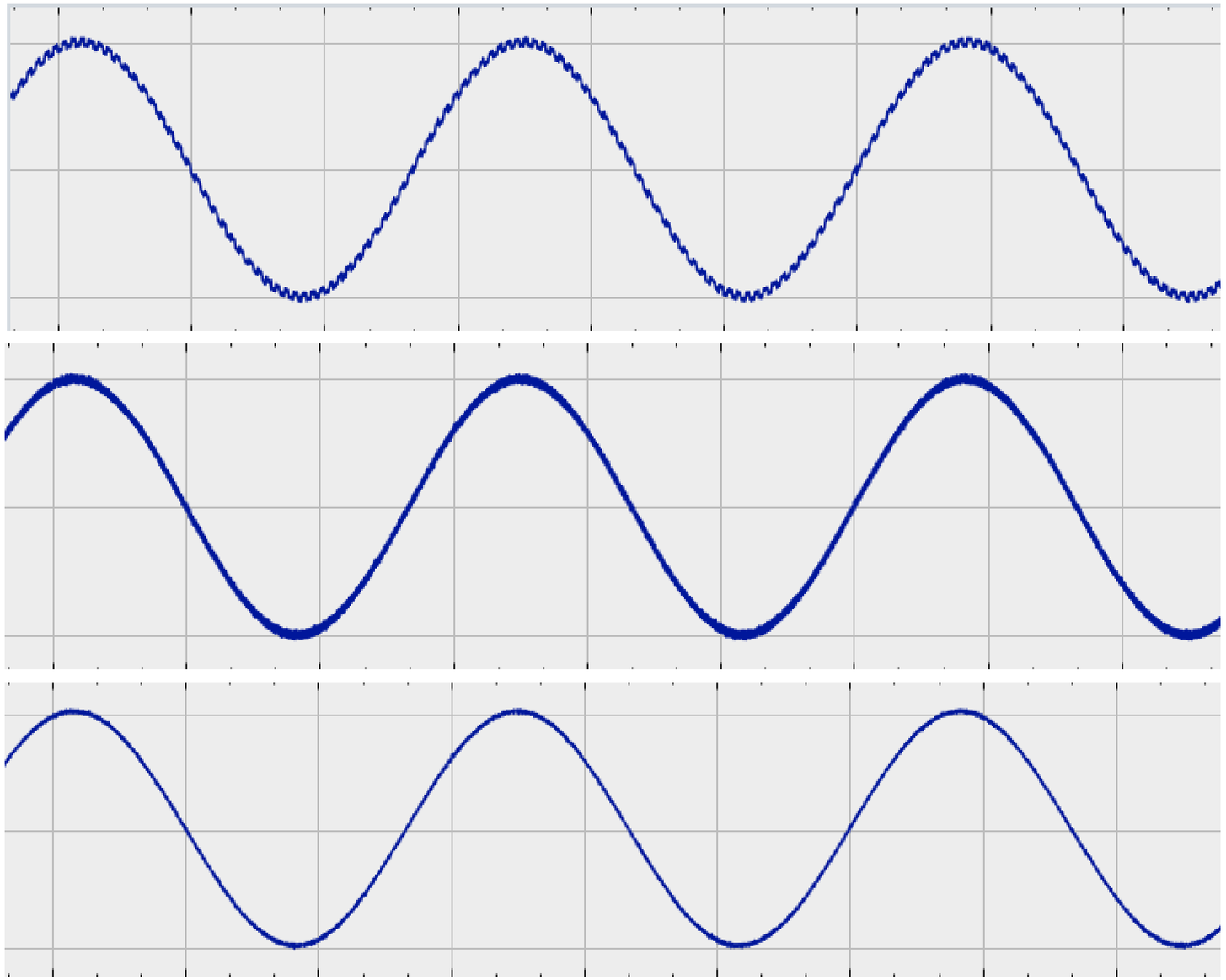


Nested-Chopping

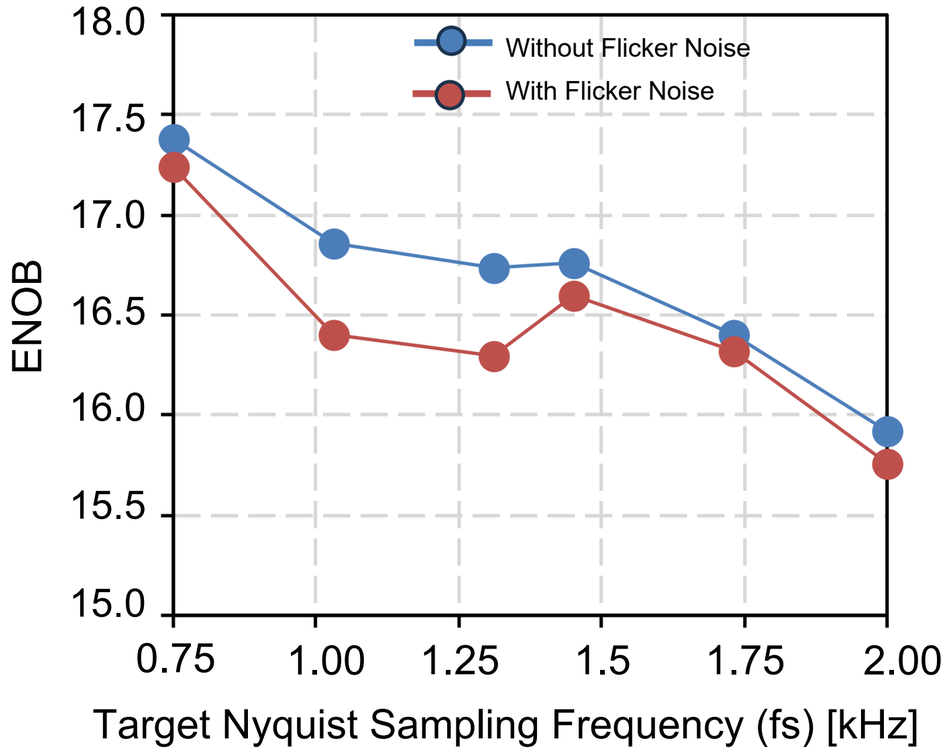
Inner Chopper: Removes 1/f Noise

Outer Chopper: Removes Residual Ripple

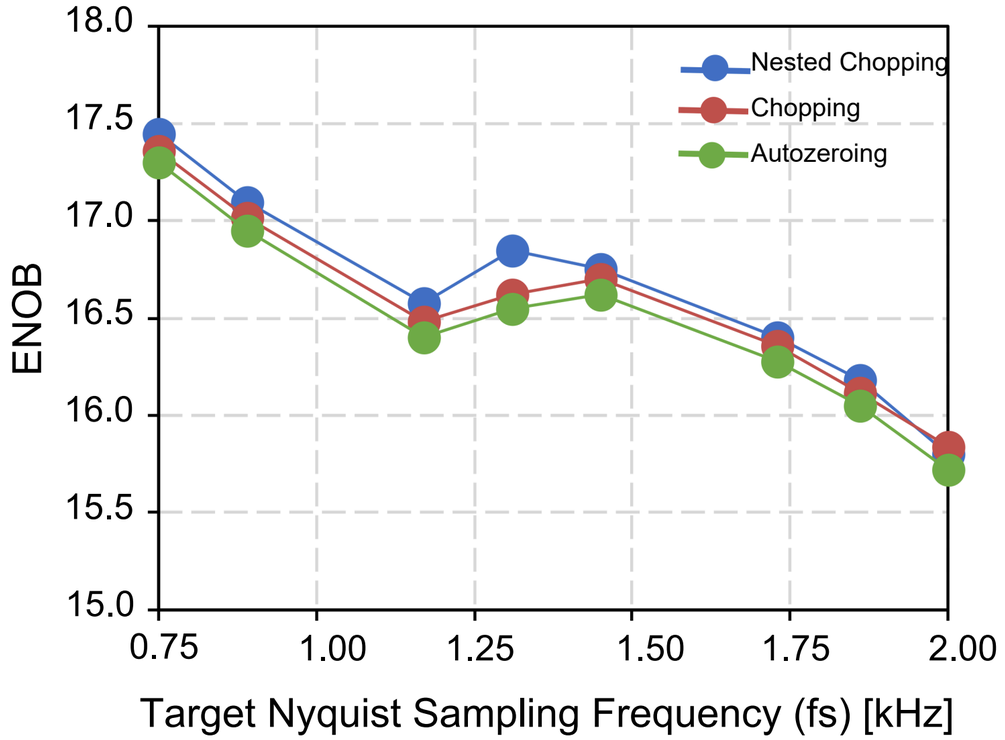
Clean Output Signal



Results and Conclusion



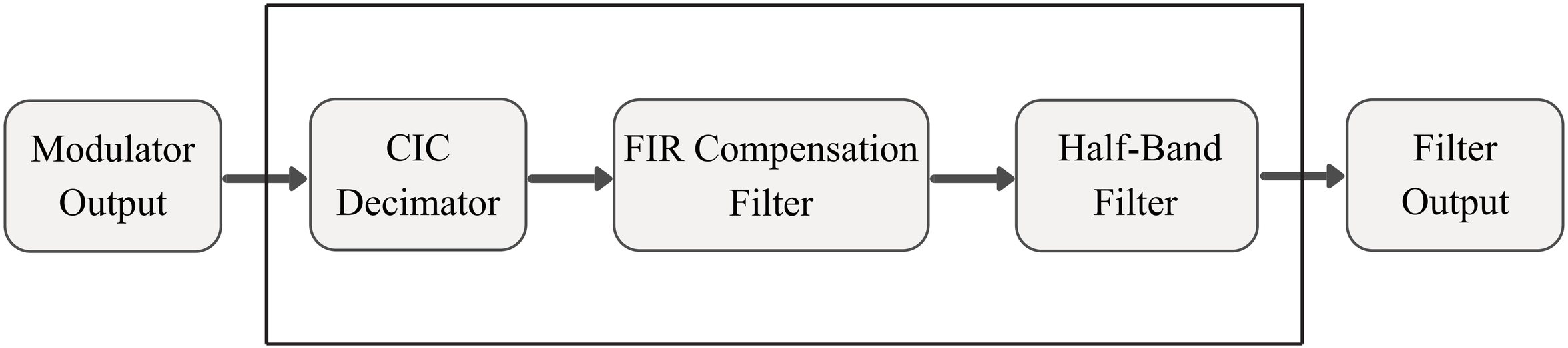
ENOB Degradation after adding Flicker noise



Improved ENOB comparison of different chopping techniques

Nested Chopping gives the cleanest, most stable low-frequency performance compared to Autozeroing and chopping

Decimation Filter



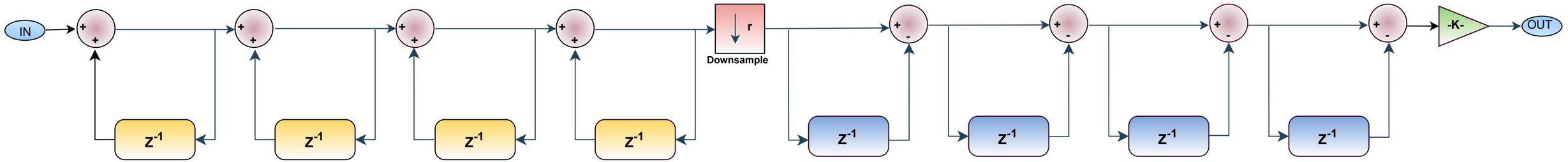
CIC Decimator

Fourth-order CIC

N = L + 1

where, N is the order of CIC,
L is the order of Modulator

Performs large decimation without using multipliers, hence is hardware efficient (Only **adders** used)



Causes **passband droop** that must be compensated

Integrator + Comb Stages

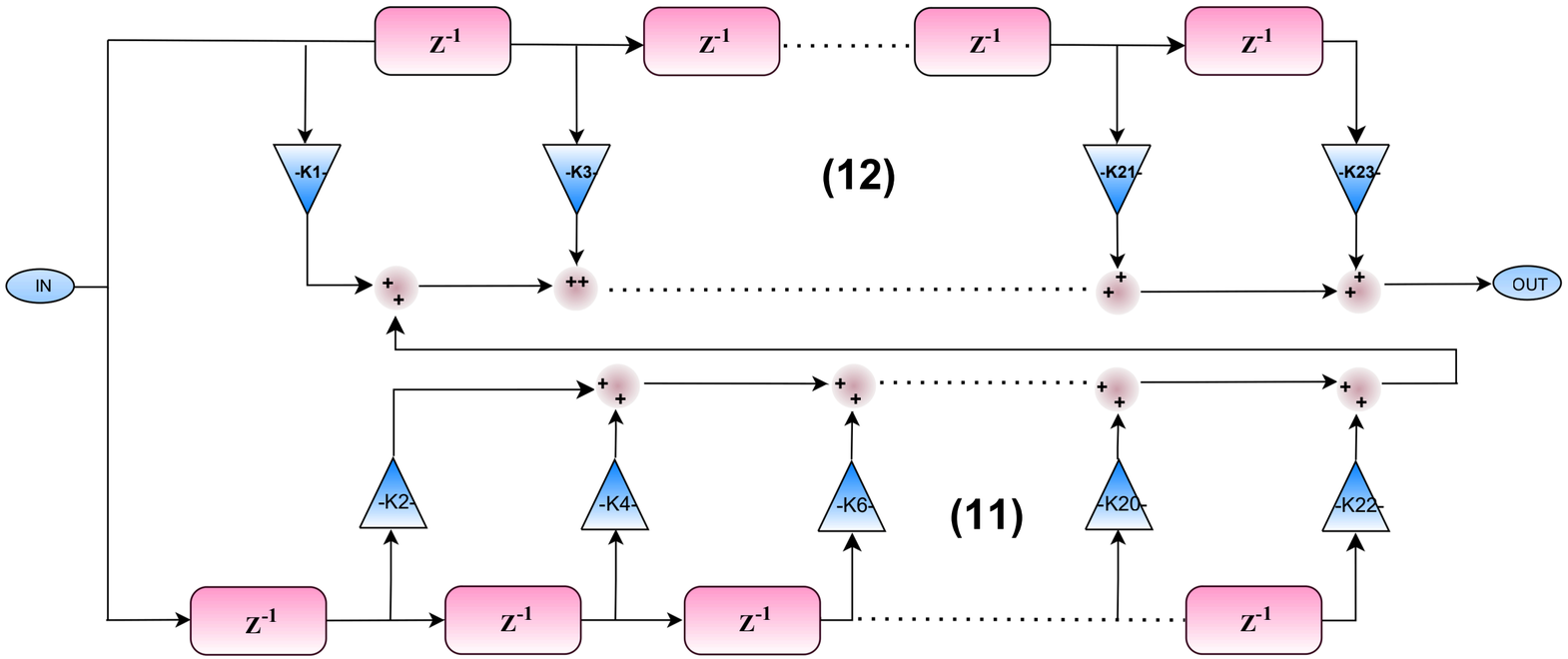
FIR Filter

FIR stage **compensates** CIC passband droop

Does the spectral shaping of the input data

Coefficients designed using **Parks-McClellan Algorithm**

Optimal performance achieved at 23 taps



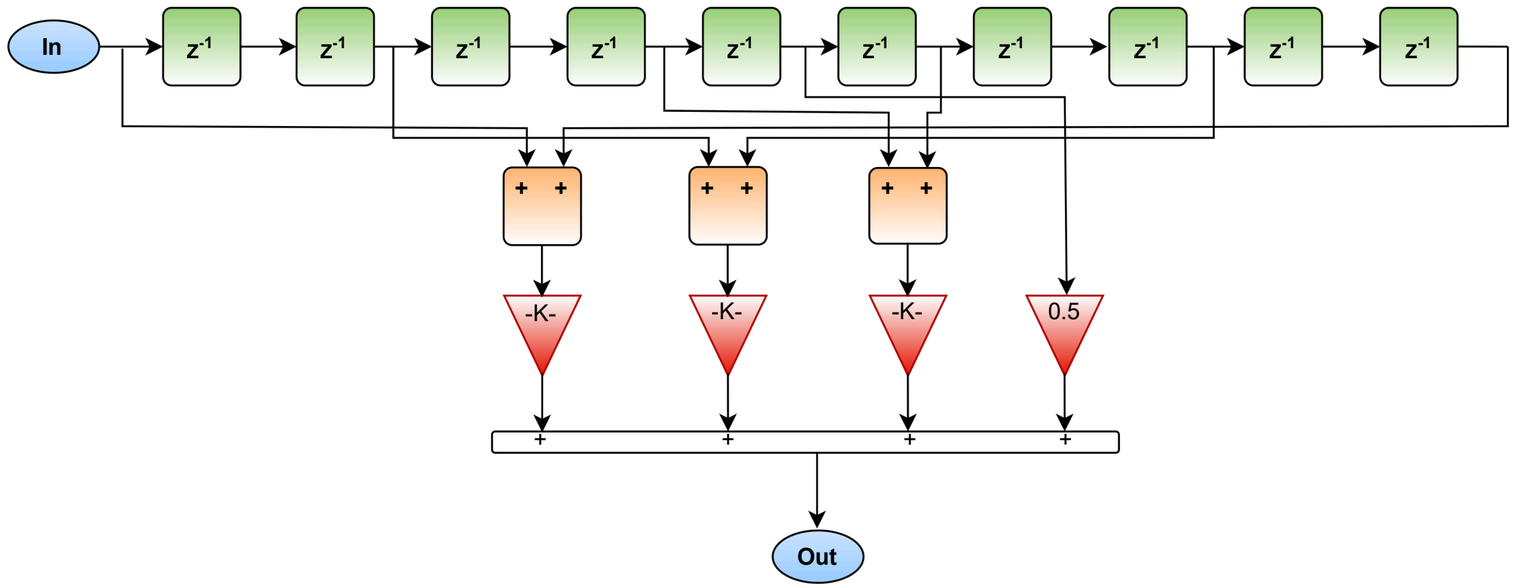
Half-Band Filter

Coefficients calculated using **firhalfband** Matlab library

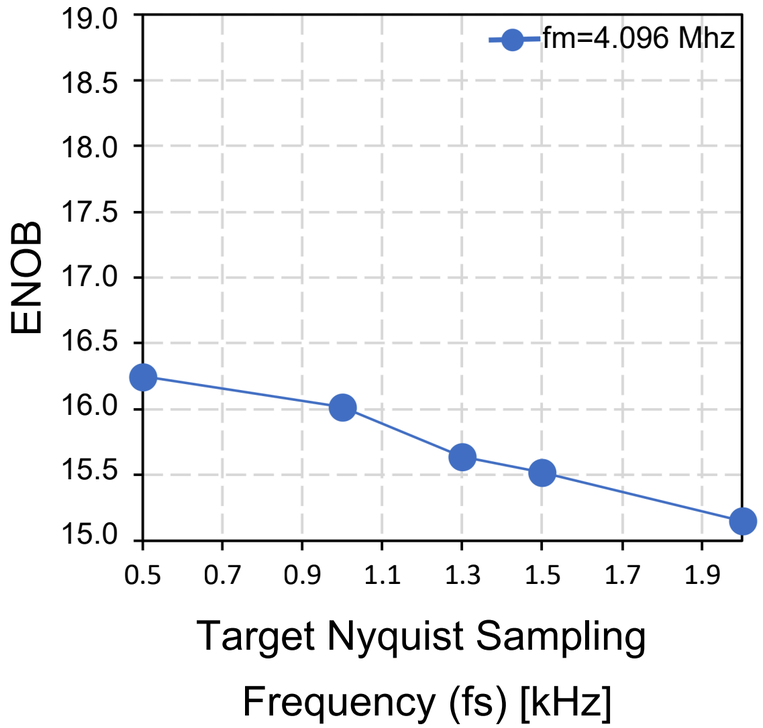
Cost-efficient when implemented, fewer adders

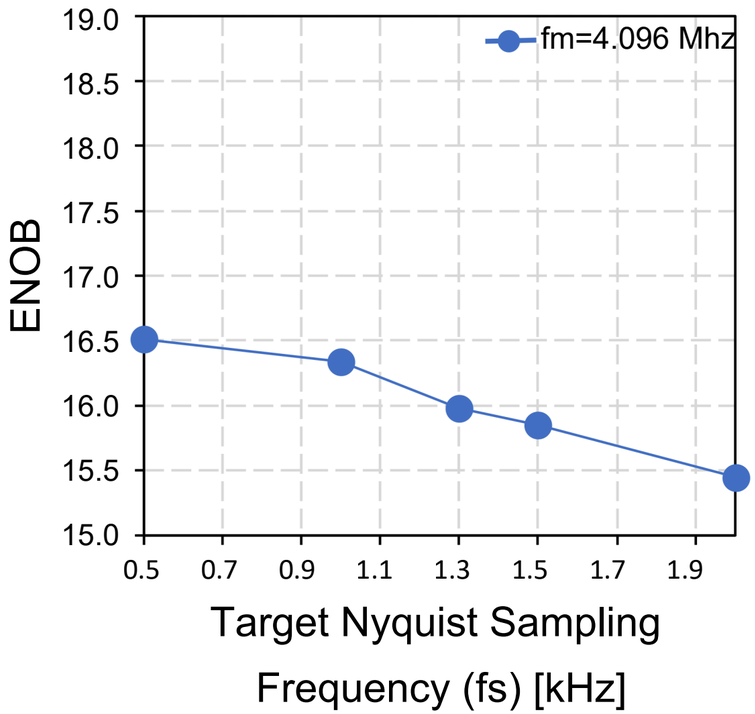
Flat passband and excellent **anti-alaising** properties

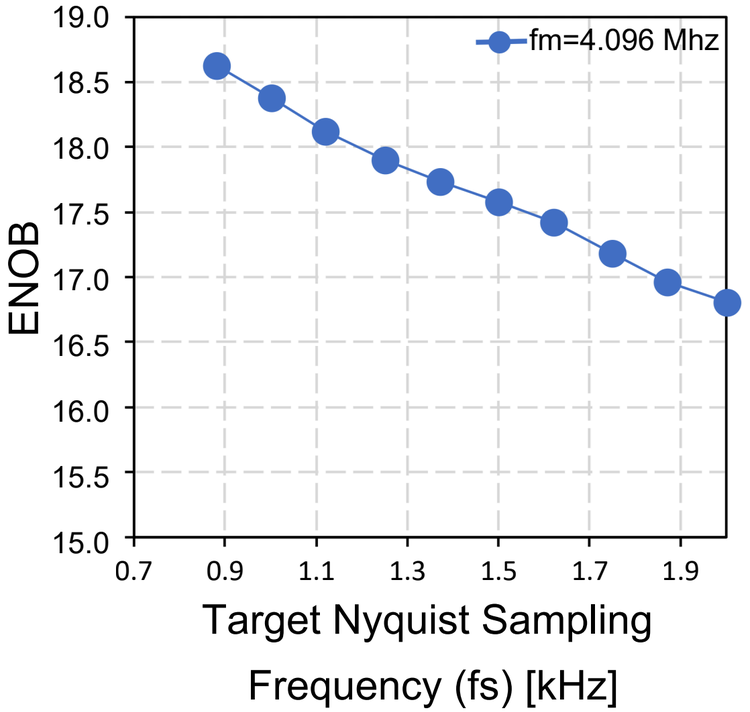
Optimal output achieved at 11 taps



Results Comparison

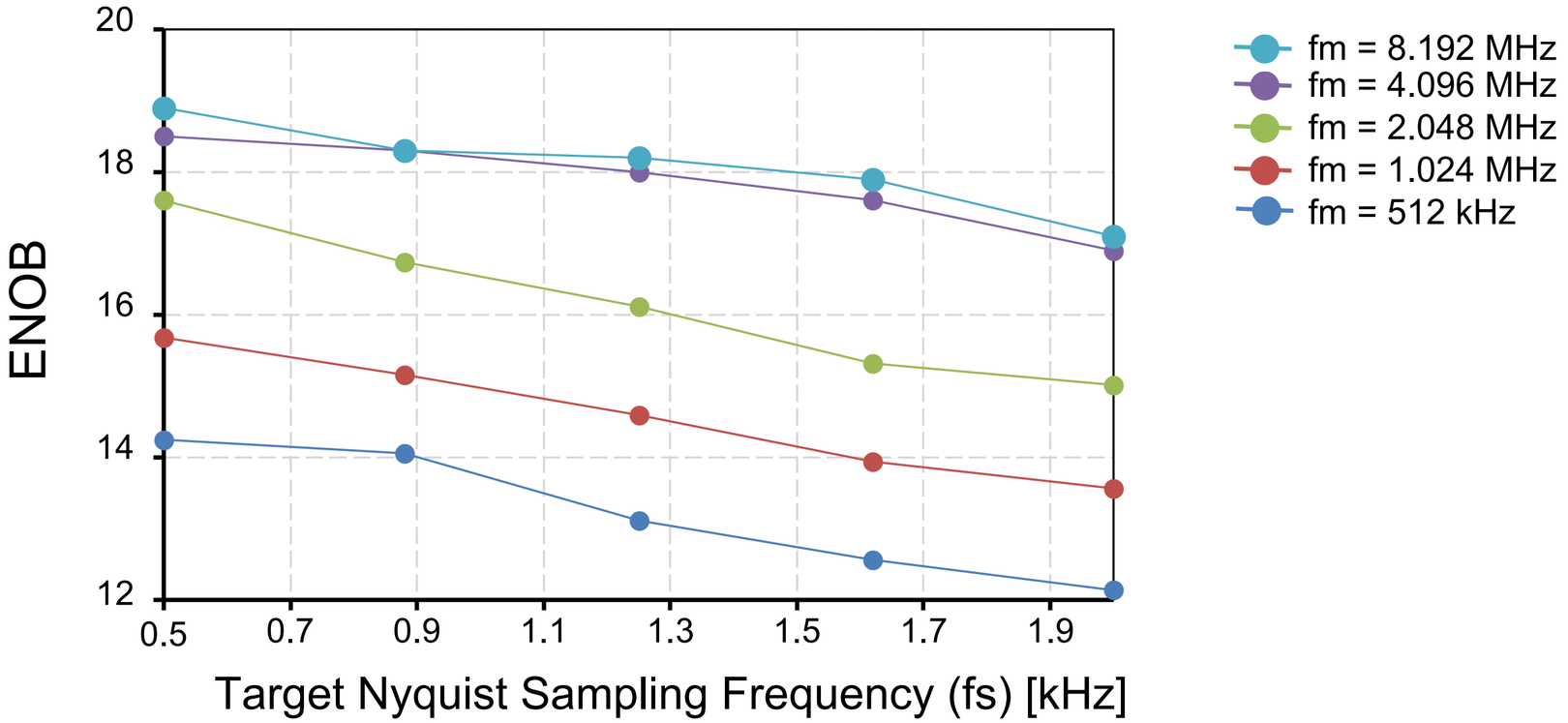






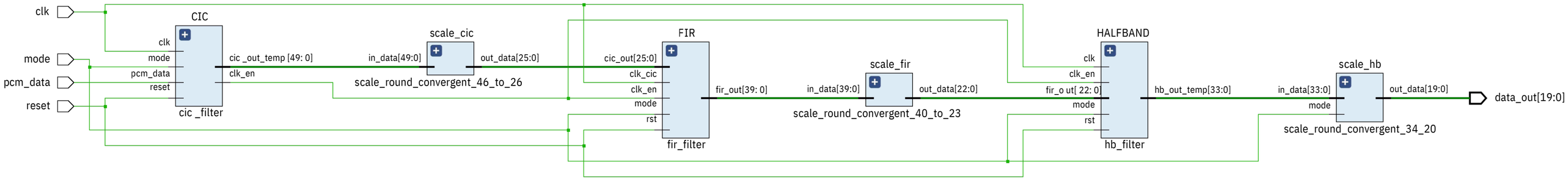
CIC OUTPUT → FIR OUTPUT → HALFBAND OUTPUT

Trade-Off



Final Obtained Result, ENOB keeps **dropping** at **higher** Nyquist Frequencies

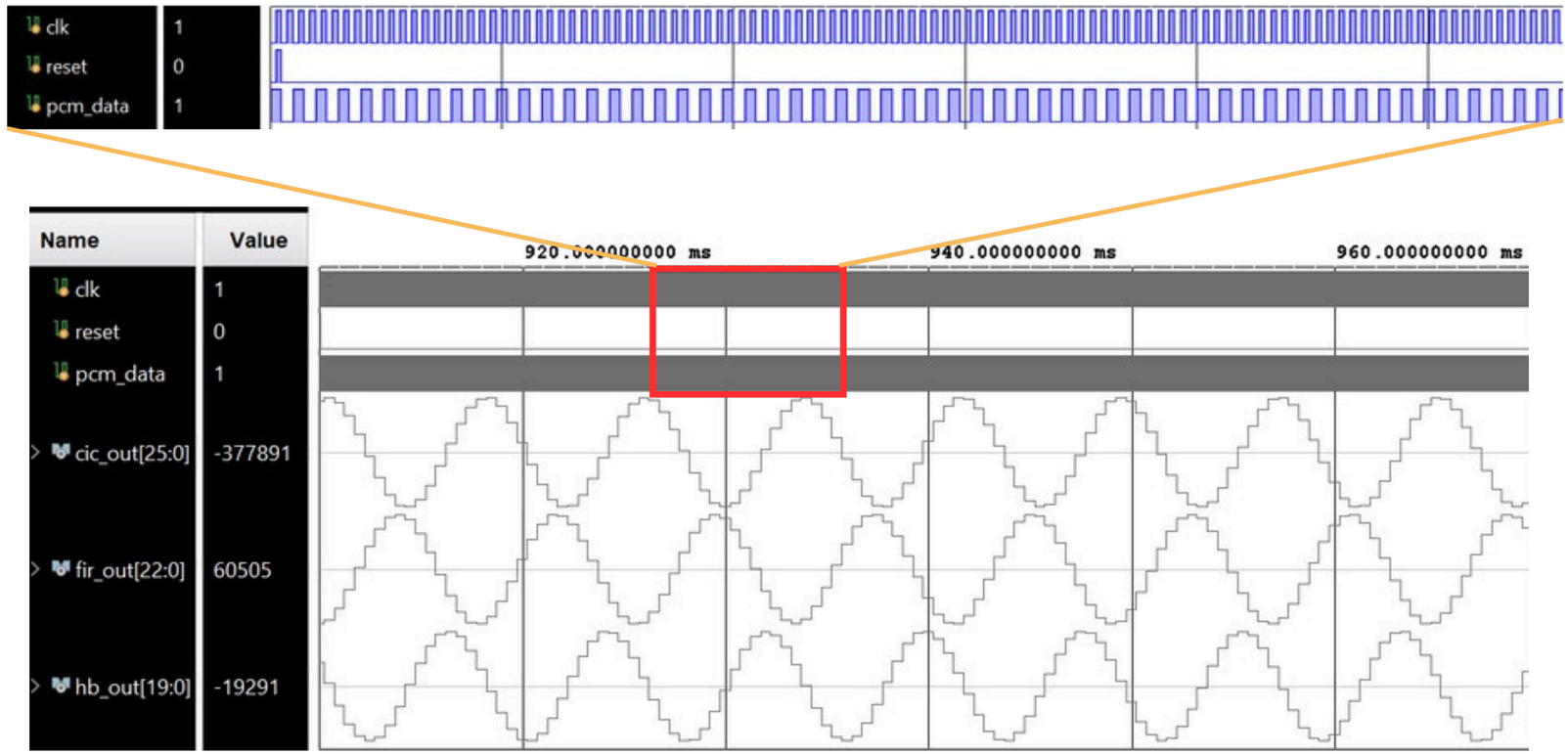
RTL Implementation



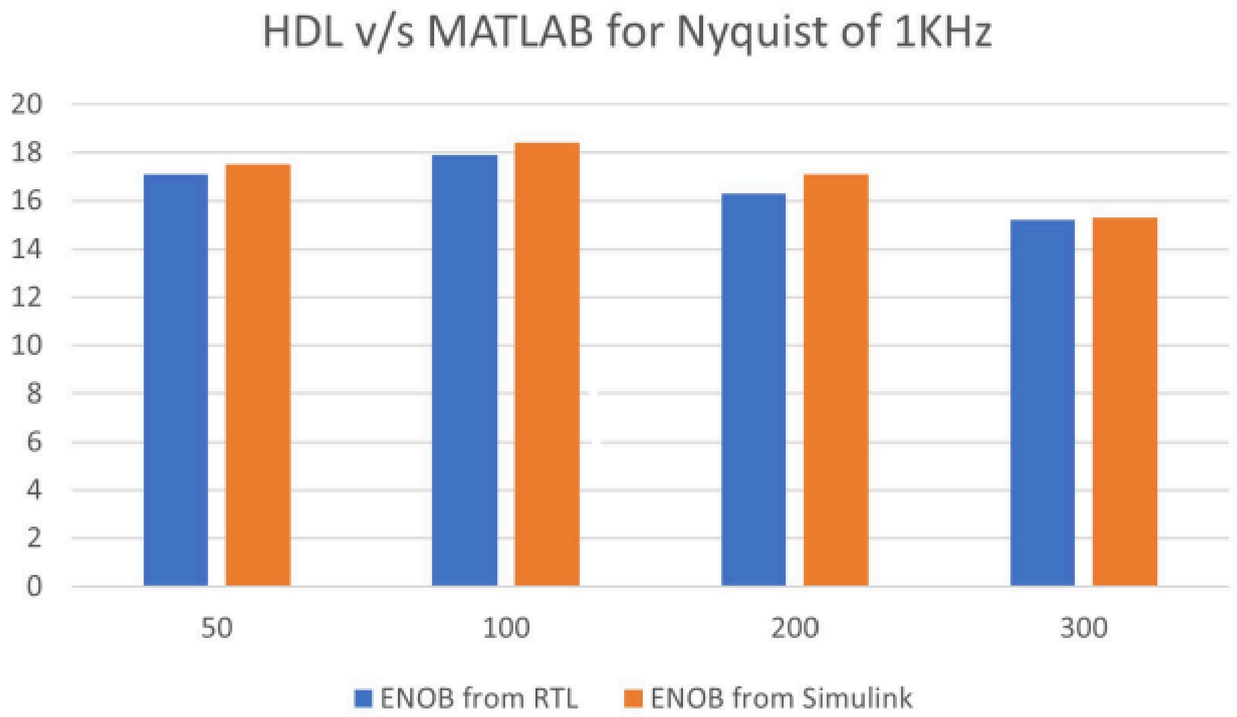
Two modes of operation:

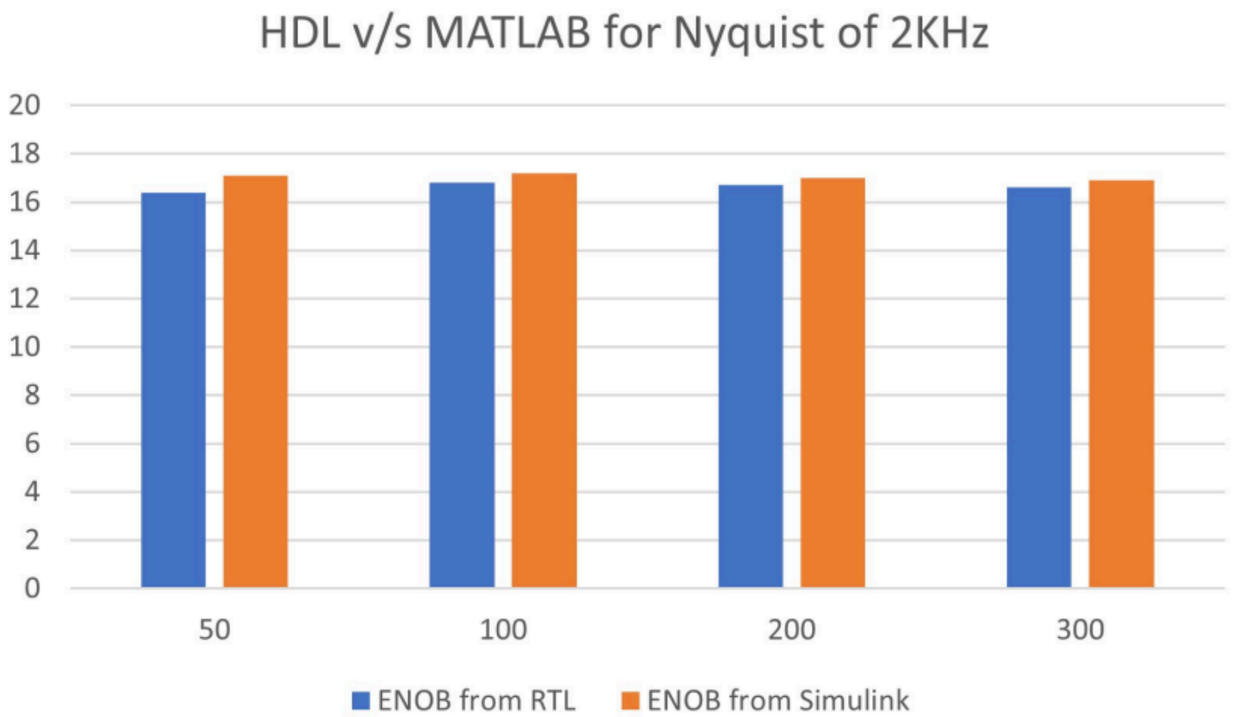
High Resolution - 1 kHz
Nyquist Rate

Low Resolution - 2 kHz
Nyquist Rate



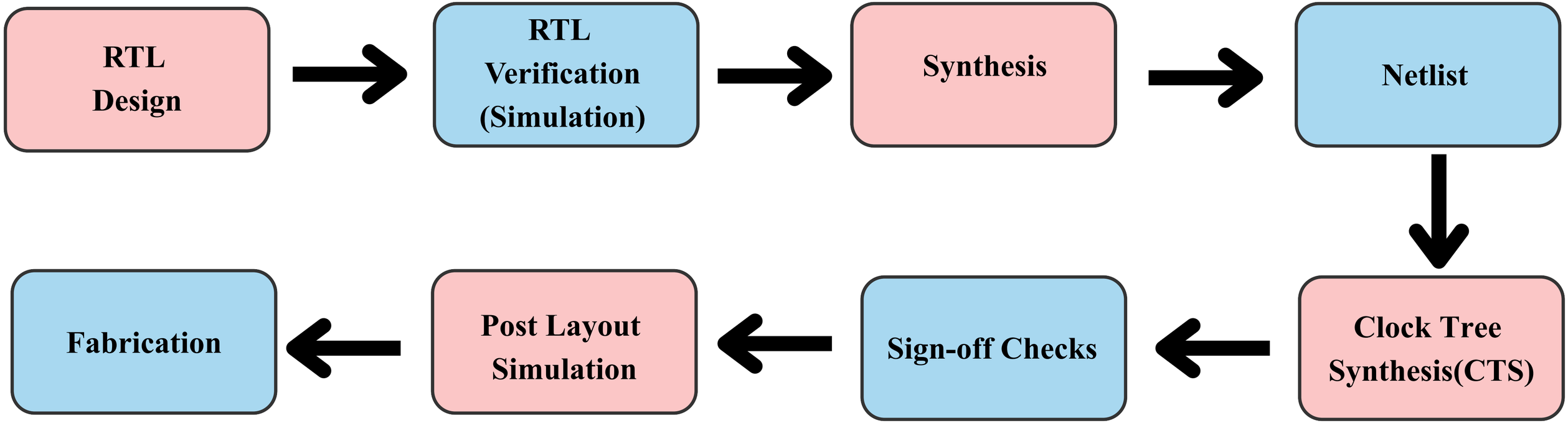
Results





The error in RTL output compared to that of Matlab was considerably small

ASIC Design Flow



Post Synthesis Area &Timing Results

Metric	Value
Total Cell Area (Cell + Physical) (mm^2)	**0.415**
Net Area (mm^2)	**0.0053**
Total Area (Cell+ Physical + Net) (mm^2)	**0.420**
Leaf Instance Count	**8,145**
Sequential Instance Count	**1,668**
Combinational Instance Count	**6,477**

Mode	Setup Time Summary	Hold Time Summary
WNS (ns)	**4.664**	**0.141**
TNS (ns)	**0**	**0**
All Paths	**4543**	**4543**
Violating Paths	**0**	**0**

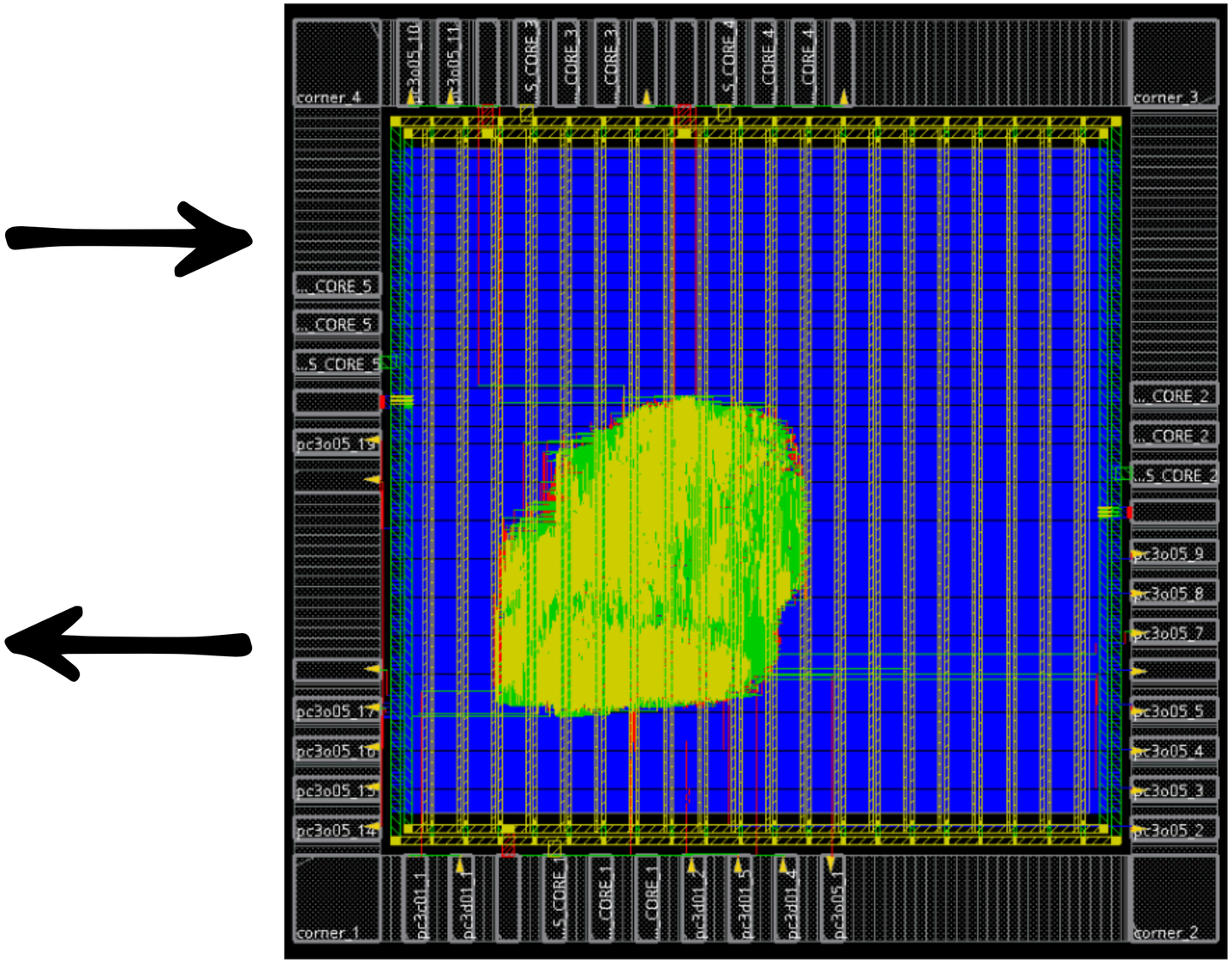
ASIC Layout

Physical Design on Cadence Innovus

Final Layout View after

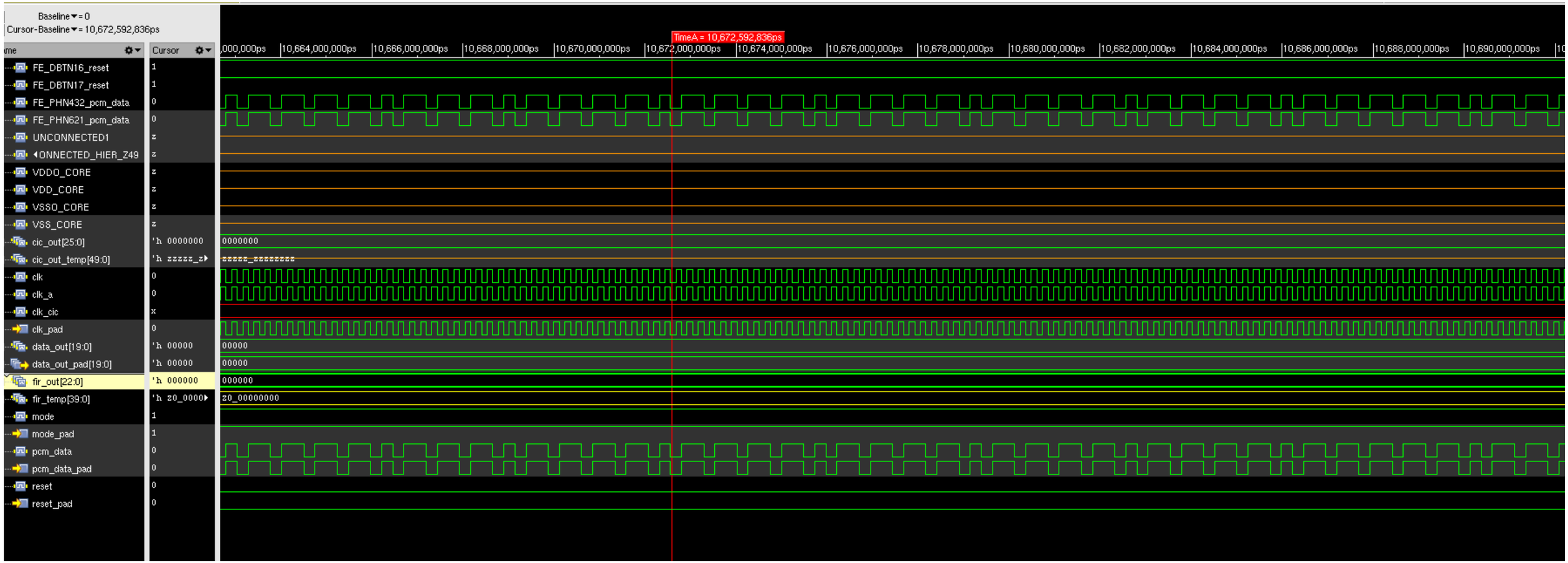
Floorplanning, Power Planning,

Standard-Cell Placement and Routing



GDSII File

Post Layout



Post layout simulation window



THANK YOU

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